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**Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology**  
**School of Computing**

**Department of Computer Science and Engineering**

**10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)**

**PROJECT REVIEW : MILESTONE PROJECT – 1**

**Date: 23-03-2026**

**Full-Stack Skill Learning Academy Marketplace with Secure Authentication**

**SDG: SDG No – SDG 4- Quality Education**

**Presented By:**

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**Courses Handling Faculty:**

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# INTRODUCTION



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- This project is a web-based platform designed to provide online skill learning opportunities for users.
- It allows users to register and login securely using authentication mechanisms like password hashing and session management.
- The system supports different roles such as Student and Instructor with role-based access control.
- Students can browse, search, and enroll in various courses available in the marketplace.
- Instructors can create, update, and manage courses and track enrolled students.
- The application is developed using HTML, CSS, JavaScript, PHP, and MySQL.
- The main goal of this project is to provide an accessible, user-friendly, and secure online learning environment.

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# PROBLEM STATEMENT



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- In today's digital era, many learners face difficulty in accessing a centralized platform that provides structured and affordable skill-based education.
- Existing systems often lack proper organization of user roles, making it difficult to manage students, instructors, and administrators efficiently.
- Many online platforms do not ensure secure authentication, leading to risks such as unauthorized access and data breaches.
- There is a need for a system that allows instructors to easily create, manage, and update courses without technical complexity.
- Traditional systems lack modular architecture, making maintenance, scalability, and feature enhancement difficult.
- There is also a need for proper database management to store and retrieve user, course, and enrollment data efficiently.
- Therefore, this project aims to develop a secure, modular, and role-based full-stack web application that addresses these challenges and provides an effective online learning marketplace.

# OBJECTIVE(S)



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- To design a responsive and user-friendly interface

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- To ensure secure data storage using MySQL database

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- To prevent SQL injection using prepared statements

- To build a modular and scalable system architect

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- To develop a full-stack web-based learning marketplace

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- To implement secure authentication using hashing and sessions

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- To provide role-based access for students and instructors

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- To enable course creation, management, and enrollment

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- To design a responsive and user-friendly interface

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- To ensure secure data storage using MySQL and prepared statements

# PROPOSED SYSTEM



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- The system provides a centralized platform for online skill learning

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- It supports secure authentication using password hashing and sessions

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- Role-based access is implemented for students, instructors, and admin

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- Users can easily browse, search, and enroll in courses

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- Instructors can create, update, and manage course content

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- The system uses a modular architecture for better scalability and maintenance

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- Data is securely stored and managed using a MySQL database

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# PROPOSED SYSTEM ARCHITECTURE



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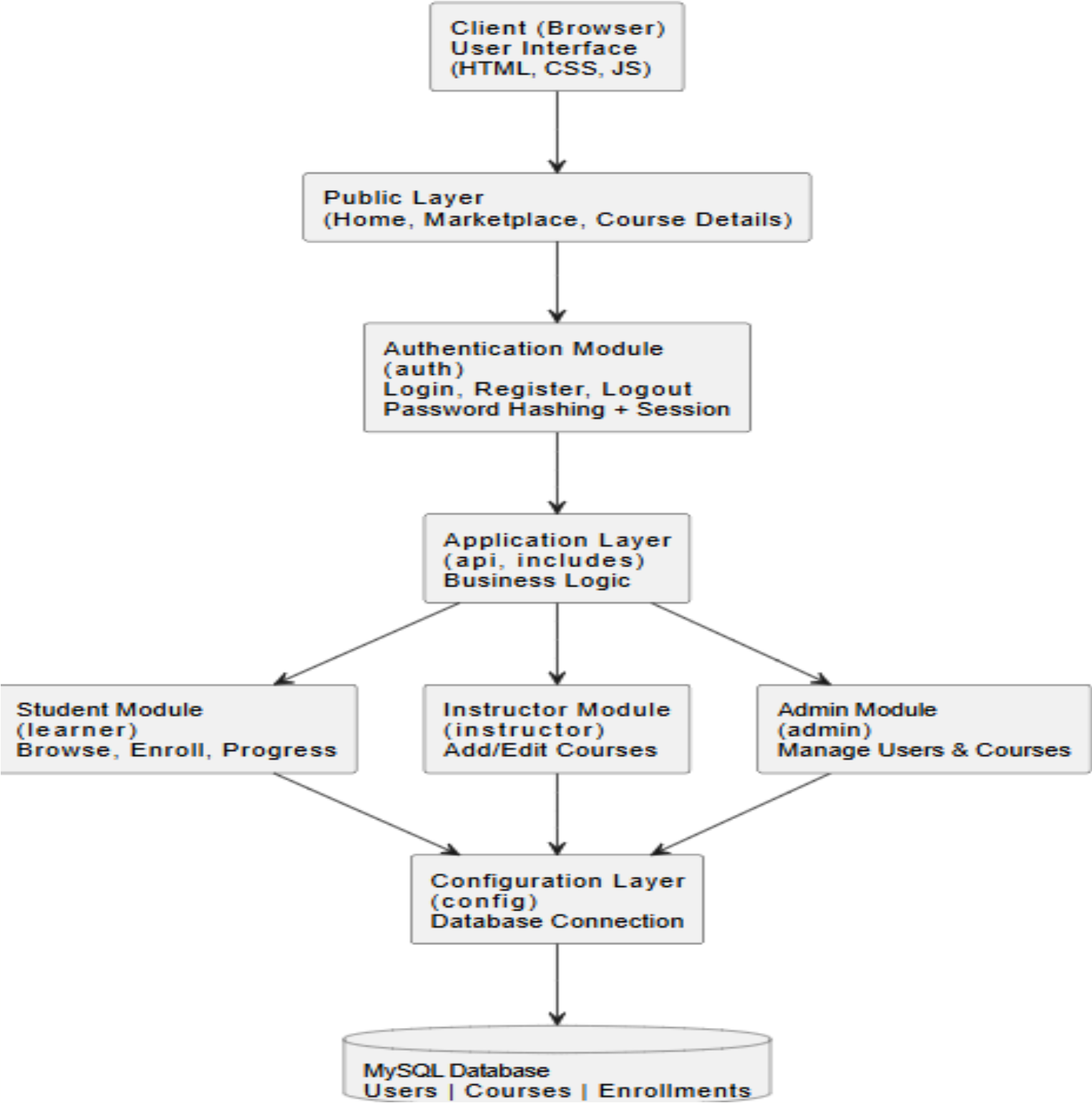
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# WORKFLOW



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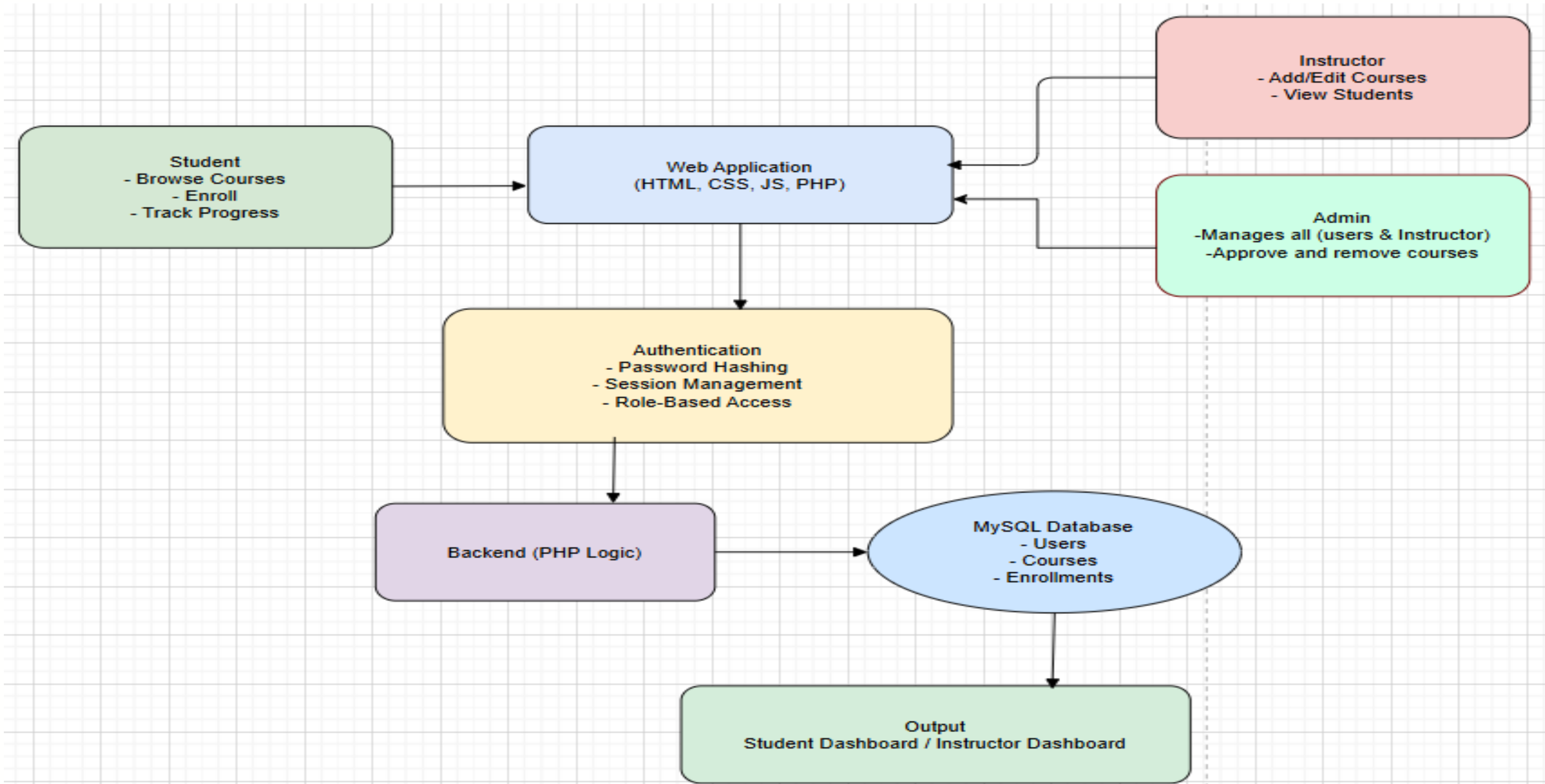
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# TECHNOLOGIES USED



## Frontend:

- HTML5
- CSS3

## Backend:

- PHP (Core PHP)
- Session Management
- MySQL(Database Connectivity)

## Database:

- MySQL
- phpMyAdmin

## Tools:

- XAMPP (Apache + MySQL Server)
- Notepad
- Web Browser (Chrome)

## Hardware / Software Requirements:

- **System:** Windows 10
- **RAM:** Minimum 4GB
- **Software:** XAMPP Server

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# MODULE DESCRIPTION



## Module 1: User Authentication Module

- 1 Handles user registration and login.
- 2 Ensures secure access using password hashing and session management.

## Module 2: Course Management Module

- 3 Allows instructors to add, update, and delete courses.
- 4 Displays available courses for students in the marketplace.

## Module 3: Enrollment Module

- 5 Enables students to enroll in courses.
- 6 Manages enrollment records and course access.

## Module 4: Learning & Progress Module

- 8 Allows students to access enrolled courses.
- 9 Tracks learning progress and completion status.

## Module 5: Dashboard Module&Module

Provides dashboards for students, instructors, and admin.  
Manages users, courses, and overall system activities.

# IMPLEMENTATION SCREENSHOTS



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## Login Page :

- Allows registered users to enter email and password to access the system
- Verifies credentials using secure authentication (password hashing & sessions)
- Redirects users to respective dashboards based on their role (student/instructor/admin)

## Register Page :

- Enables new users to create an account by entering name, email, and password.
- The system stores the user details securely in the database
- After successful registration, the user can login and access the academy marketplace.

# IMPLEMENTATION SCREENSHOTS



## Exam page:

- Displays available courses and allows students to select and enroll
- Provides course content and conducts assessments/exams for learning evaluation
- Records student performance and updates progress based on results

Course Test: Introduction to Python Programming

1. Which keyword defines a function in Python?

☐ A. func

☐ B. define

☐ C. def

☐ D. fn

2. What is the output of `print(2 + 3 * 2)`?

☐ A. 10

☐ B. 12

☐ C. 8

☐ D. 7

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8(A)

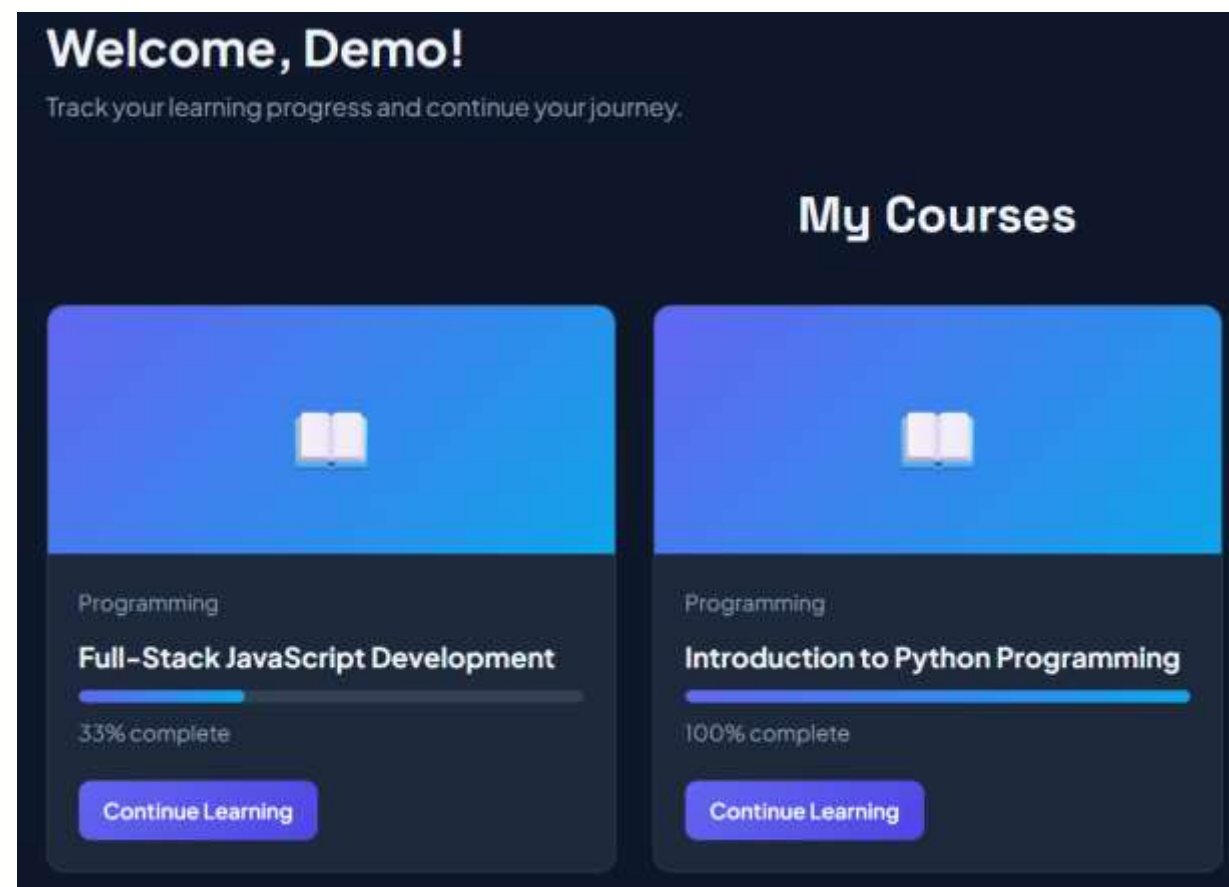
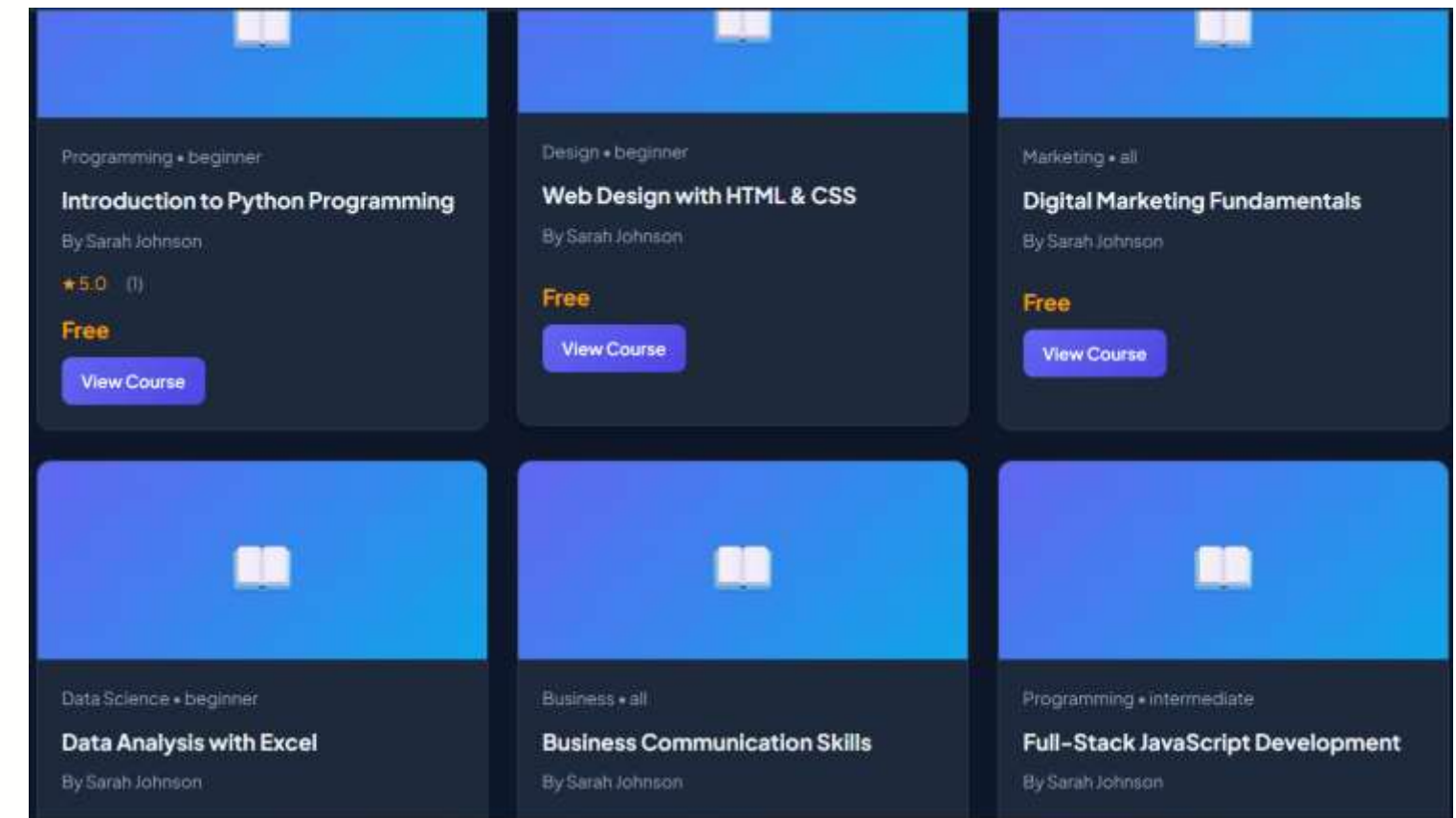
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# IMPLEMENTATION SCREENSHOTS



## Course Marketplace

- Displays available courses with details like title, description, and price
- Allows users to search, filter, and select courses for enrollment



## My Courses

- Displays all courses enrolled by the student in one place
- Allows users to access course content and track their learning progress

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# IMPLEMENTATION SCREENSHOTS



My Courses + Add Course

| Course                             | Category     | Students | Status    | Actions                                         |
|------------------------------------|--------------|----------|-----------|-------------------------------------------------|
| Introduction to Python Programming | Programming  | 1        | Published | <button>Edit</button> <button>Students</button> |
| Web Design with HTML & CSS         | Design       | 0        | Published | <button>Edit</button> <button>Students</button> |
| Digital Marketing Fundamentals     | Marketing    | 0        | Published | <button>Edit</button> <button>Students</button> |
| Data Analysis with Excel           | Data Science | 0        | Published | <button>Edit</button> <button>Students</button> |
| Business Communication Skills      | Business     | 0        | Published | <button>Edit</button> <button>Students</button> |
| Full-Stack JavaScript Development  | Programming  | 1        | Published | <button>Edit</button> <button>Students</button> |
| UI/UX Design Masterclass           | Design       | 0        | Published | <button>Edit</button> <button>Students</button> |

## My Courses (Instructor):

- Displays all courses created by the instructor
- Allows instructors to edit, update, or delete their courses

## Admin Dashboard

- Provides overview of users, courses, and system activities
- Allows admin to manage users, courses, and platform operations

|                              |                |             |
|------------------------------|----------------|-------------|
| Total Users                  | Total Courses  | Enrollments |
| 3                            | 10             | 2           |
| Revenue                      |                |             |
| \$49.99                      |                |             |
| Recent Users                 | Recent Courses |             |
| Name                         | Role           | Joined      |
| Demo Learner                 | learner        | Mar 2, 2026 |
| Platform Admin               | admin          | Mar 2, 2026 |
| Sarah Johnson                | instructor     | Mar 2, 2026 |
| Title                        | Instructor     | Status      |
| Startup & Entrepreneurship   | Sarah Johnson  | Published   |
| Machine Learning with Python | Sarah Johnson  | Published   |



# ADVANTAGES & LIMITATIONS



## Advantages

- Provides a centralized online learning platform
- Secure authentication ensures data safety
- Role-based access improves system control
- Easy course management for instructors
- User-friendly and responsive interface
- Scalable and modular architecture

## Disadvantages

- Requires internet connection to access
- Basic security (not fully industry-level)
- Depends on server performance (XAMPP/local)
- Limited features compared to large platforms
- Maintenance required for updates

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***Thank you***